Lecture 10 Compendium: GIS for Pest and Disease Monitoring

Module 3: Technologies of GIS in Crop Protection

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Compendium for Advanced Study

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1 Introduction: Spatial Epidemiology

Epidemiology is the study of disease patterns. **Spatial Epidemiology** adds the dimension of "Where" and "Why there?".

In agriculture, pests and diseases do not appear randomly. They follow specific spatial processes driven by environmental covariates (wind, humidity, host density). At the PhD level, we move beyond simply mapping "occurrence points" to modeling the underlying **stochastic processes**.

1.1 The Spatial Disease Triangle

The classical Disease Triangle (Host, Pathogen, Environment) must be spatialized:

- **Host:** Where is the crop? (Land Use Map).
- **Pathogen:** Where is the inoculum? (Point occurrences).
- **Environment:** What are the local conditions? (Interpolated Climate Surfaces).

Risk is the spatial intersection of these three layers.

$$\text{Risk}(x, y) = f(\text{Host}(x, y) \cap \text{Pathogen}(x, y) \cap \text{Environment}(x, y))$$

2 Point Pattern Analysis: Testing for Randomness

Before we can claim a "hotspot" exists, we must statistically prove that the observed pattern is not random.

2.1 Complete Spatial Randomness (CSR)

The null hypothesis (H_0) states that pests are distributed according to a Poisson process (randomly) across the study area.

2.2 1. Ripley's K-Function

A multi-scale analysis. It counts the number of neighbors within a distance d of a random point.

$$K(d) = \lambda^{-1} E$$

Where E is the expected number of extra events within distance d of a randomly chosen event, and λ is the density (events per unit area).

- If $L(d) > 0$: **Clustered** (Contagion/Infection process).
- If $L(d) < 0$: **Dispersed** (Competition process).

2.3 2. Moran's I (Spatial Autocorrelation)

Measures whether similar values are clustered together.

$$I = \frac{N}{\sum_i \sum_j w_{ij}} \frac{\sum_i \sum_j w_{ij}(x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2} \tag{1}$$

- $I \approx +1$: High values cluster with high values (Hotspot).
- $I \approx -1$: High values cluster with low values (Checkerboard).
- $I \approx 0$: Random.

3 Hotspot Mapping: Kernel Density Estimation (KDE)

Dots on a map are hard to interpret. KDE converts discrete points into a continuous surface of **Probability Density**.

3.1 The Math of KDE

For each point x_i, we place a "Kernel" (a 3D bump, usually Gaussian). We sum all bumps to get the surface.

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

Where:

- K: The kernel function (Gaussian).
- h: The **Bandwidth** (Search Radius). This is the most critical parameter.
- n: Number of points.

PhD Research Tip

Bandwidth Selection: Choosing a small h results in a "spiky" map (overfitting). Choosing a large h results in an overly smooth map (underfitting). Use statistical methods like "Silverman's Rule of Thumb" to optimize h.

3.2 Python: Generating a KDE Heatmap

Simulating a pest outbreak and generating a heatmap.

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import gaussian_kde

# 1. Generate Mock Data (Pest Locations)
# Two clusters of infestation (Gaussian distribution)
cluster1 = np.random.normal(loc=[2, 2], scale=0.5, size=(50, 2))
cluster2 = np.random.normal(loc=[6, 6], scale=1.0, size=(50, 2))
background = np.random.uniform(0, 10, size=(20, 2)) # Random background noise

points = np.vstack([cluster1, cluster2, background])
x, y = points.T

# 2. Calculate KDE
# Grid for evaluation
xi, yi = np.mgrid[0:10:100j, 0:10:100j]
positions = np.vstack([xi.ravel(), yi.ravel()])
values = np.vstack([x, y])

kernel = gaussian_kde(values)
# Bandwidth (Scott's Rule is default in scipy)
zi = np.reshape(kernel(positions).T, xi.shape)

# 3. Visualize
plt.figure(figsize=(8, 6))
# Plot surface
plt.pcolormesh(xi, yi, zi, shading='auto', cmap='Reds')
# Plot original points
plt.scatter(x, y, s=5, c='black', alpha=0.5, label='Infestation Point')
```

```
plt.colorbar(label='Risk Density')
plt.title('Pest Risk Heatmap (KDE)')
plt.legend()
plt.show()
```

Listing 1: Kernel Density Estimation with Python

4 Predictive Modeling: Species Distribution Models (SDM)

We don't just want to know where pests *are*; we want to know where they *could be*.

4.1 Maximum Entropy (MaxEnt)

MaxEnt is the standard machine learning algorithm for presence-only data (we know where the pest is, but we rarely have confirmed "absence" data).

The Logic: Find the probability distribution of maximum entropy (most uniform) subject to the constraints imposed by the observed environmental variables at the presence locations.

Example: If Fall Armyworm is found only where Temp $>$ 25řC and Rainfall $<$ 500mm, MaxEnt identifies all other locations in Tanzania matching these criteria as "High Risk."

4.2 Covariates for Pest Modeling

For your PhD models, download these datasets:

1. **Bioclimatic Variables (WorldClim):** 19 variables (e.g., Bio1 = Annual Mean Temp, Bio12 = Annual Precip).
2. **Topography (SRTM):** Elevation (affects temperature) and Slope.
3. **Vegetation (MODIS/Sentinel):** NDVI (Host availability).

5 Spatio-Temporal Dynamics

Pests move. Static maps are insufficient.

5.1 The Space-Time Cube

A visualization technique where X and Y are geography, and Z is Time. * **Vertical columns:** Persistent hotspots (Endemic). * **Diagonal lines:** Moving hotspots (Epidemic spread/Migration).

5.2 Wind Dispersion Modeling

For airborne pathogens (e.g., Wheat Rust spores), distance is not Euclidean. It is **Anisotropic** (direction-dependent). We model the **Effective Distance** based on wind vector fields.

$$P(x, y, t) = \text{Source} + \text{Diffusion} + \text{Advection (Wind)}$$

6 Case Study: Fall Armyworm (FAW) in Tanzania

Context: FAW (*Spodoptera frugiperda*) is a transboundary pest. **GIS Intervention:** 1. **Data Collection:** Farmers use a mobile app (Push-Pull) to report sightings. GPS coordinates are uploaded. 2. **Analysis:** * **KDE:** Identifying current outbreaks. * **MaxEnt:** Predicting future spread based on Maize phenology (NDVI) and Wind speed. 3. **Action:** SMS alerts sent to farmers in the "High Probability" path 2 weeks before arrival.

7 Review Questions & Assignment

1. **Statistics:** You perform a Moran's I test on a dataset of Whitefly counts, and the result is $I = -0.8$ ($p < 0.05$). What does this spatial pattern look like? What biological process might cause this?

2. **Modeling:** Why is MaxEnt preferred over Logistic Regression for pest modeling when you rely on "Citizen Science" data (e.g., farmers taking photos)?

3. **Design:** Design a sampling strategy to validate a satellite-based Locust Risk Map. How do you ensure you test for "False Negatives"?

References

1. Phillips, S. J., et al. (2006). *Maximum entropy modeling of species geographic distributions.* Ecological Modelling.

2. ESRI. *How Spatial Autocorrelation (Moran's I) Works.* ArcGIS Pro Documentation.

3. WorldClim Database. `https://www.worldclim.org/`